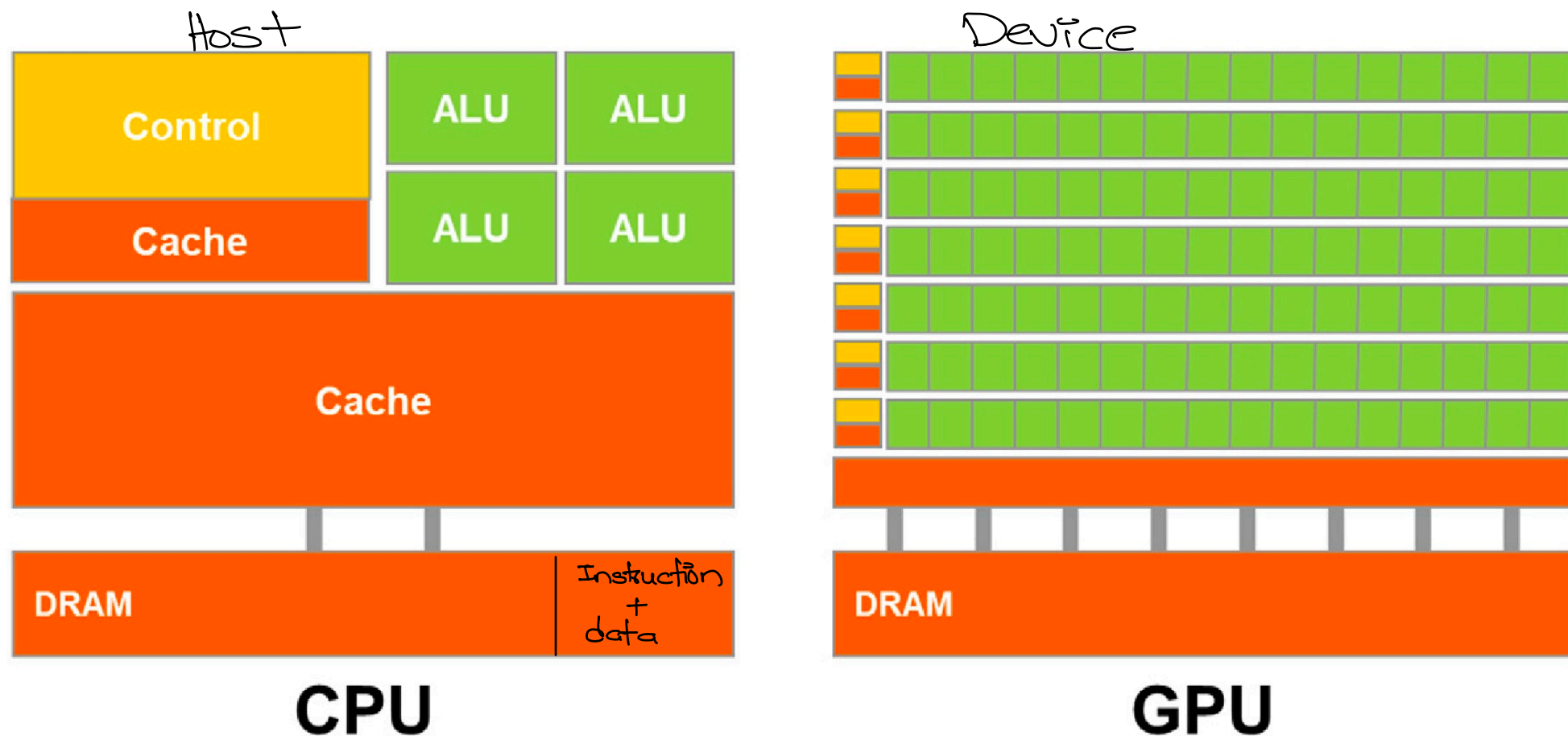




Program is first loaded in CPU's RAM & first run on CPU.

To run on GPU, CPU tells GPU to do work. This happens implicitly i.e. programmer doesn't handle this.

Programmer copies relevant data from CPU's RAM to GPU's RAM.

This is an example of array processors studied in CA.

Vector processors, on the other hand, have large registers into which entire data/array is loaded.

Typically, 512 bit & are part of the CPU.

A kernel is a function called by CPU to run on GPU.

Arrays are arranged in row-major in RAM.

So,

For 2D array index is computed as:

$$i \times W + j$$

For 3D array:

$$i \times (H \times W) + j \times W + k$$

